

Source: docs/FUTURE_HARDWARE.md • Status: Verified & Synchronized

Status: FUTURE / OUT OF SCOPE FOR V1

This specification documents forward-looking architectural designs and communication protocols for an autonomous physical robotic solver rig planned for Cubyntra V4. No physical robotics hardware or drivers are part of the Cubyntra V1 release.

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2. Microcontroller & Driver Hardware Architecture

2.1 Compute Unit: ESP32-S3

- Dual-Core Xtensa LX7 @ 240 MHz:
- Core 0: High-speed Web Serial / BLE communications and command parsing.
- Core 1: Hardware timer interrupts generating step pulses up to 100 kHz.
- Onboard Flash: 8 MB for offline move caching.

2.2 Motor Drivers: TMC2209

- StealthChop2 / SpreadCycle: Ultra-quiet operation and high torque output during rapid accelerations.
- Microstepping: 1/16 microstepping with interpolation to 1/256 for smooth positioning.
- StallGuard4: Sensorless stall detection used for automatic home calibration and physical jam detection.

3. Communication Protocol (Web Serial / BLE)

3.1 Packet Specification

Commands are transmitted as lightweight, deterministic JSON or binary packets over standard Web Serial at 115,200 baud.

JSON Representation:

```
{
  "seq": 104,
  "cmd": "EXECUTE_SOLVE",
  "moves": [
    { "face": "R", "turns": 1, "speed_rpm": 450 },
    { "face": "U", "turns": -1, "speed_rpm": 450 },
    { "face": "F", "turns": 2, "speed_rpm": 600 }
  ]
}
```

Compact Binary Protocol (4 bytes per move):

```
[ Byte 0: Face ID ('U' = 0x01, 'D' = 0x02, 'F' = 0x03, 'B' = 0x04, 'L' = 0x05, 'R' = 0x06) ]
[ Byte 1: Rotation Direction (0x01 = +90°, 0xFF = -90°, 0x02 = 180°) ]
[ Byte 2: Acceleration Profile ID ]
[ Byte 3: CRC8 Checksum ]
```

4. Kinematics & Mechanical Safety Considerations

4.1 S-Curve Acceleration Profiling

Step motors undergoing instantaneous velocity changes induce mechanical shock, leading to piece popping or corner twisting:

- Jerk-limited third-order polynomial S-curve velocity profiles:
$$j(t) = \frac{da(t)}{dt} = \text{constant}$$
- Maximum rotational velocity: 600 RPM (100ms per 90° turn).

4.2 Interlock & Collision Prevention

A physical Rubik's cube cannot execute perpendicular face turns simultaneously (e.g., R and U cannot rotate at the exact same instant without collision):

- The firmware scheduler enforces a minimum 15ms inter-move dwell time.
- Opposite face turns (e.g., R and L, or U and D) may be executed concurrently to optimize physical solve time.

4.3 Position Feedback & Error Recovery

Each stepper axis is equipped with an AS5600 12-bit magnetic rotary encoder on the output shaft:

- Confirms precise $\pm 0.5^\circ$ angular alignment at the completion of every move.
- In the event of a mechanical bind, motion halts instantly, reporting an emergency stop packet to the Cubyntra web client.